15 · Detecting and treating genetic disease

The words of this track, as it says them. Bold brown is the examiner.

Detecting and treating genetic disease. Four marks.

Explain some of the ways genetic abnormalities are detected, including genetic testing and fetal testing, and what genetic engineering and gene therapy are.

The hook: look, sample, change, treat. Look at the DNA. Sample the baby. Change the DNA. Treat the patient.

Look. Genetic testing reads a person's DNA, often from a blood sample, for genes or variants that cause disease. For example, a karyotype, which looks at the chromosomes. Or a molecular karyotype, called a CGH array, which checks thousands of positions for missing or extra DNA.

Sample the baby. There are two fetal tests. Chorionic villus sampling, or CVS: tissue is taken from the placenta, at eight to twelve weeks. And amniocentesis: amniotic fluid is taken, at fourteen to twenty weeks. In both, the fetal cells are examined for chromosomal and genetic disorders.

Change. Genetic engineering is the deliberate alteration of DNA, to change an organism's genetic make-up. Her example: bacteria altered to make insulin.

Treat. Gene therapy is the treatment of a genetic disease by inserting genetic material into the patient's cells, carried in by a vector.

The traps. CVS is earlier, eight to twelve weeks, from the placenta. Amniocentesis is later, fourteen to twenty weeks, from the fluid. Second: genetic engineering changes DNA in any organism, while gene therapy treats a patient. Third: a karyotype shows the number and shape of the chromosomes. It can't see a single-gene change like Huntington's.

Once more. Detecting and treating genetic disease.

Genetic testing reads a person's DNA, often from a blood sample: a karyotype, or a CGH array for missing or extra DNA. Chorionic villus sampling takes placenta at eight to twelve weeks. Amniocentesis takes amniotic fluid at fourteen to twenty weeks. Both check the fetal cells. Genetic engineering is the deliberate alteration of DNA, like bacteria altered to make insulin. Gene therapy treats a genetic disease by inserting genetic material into the patient's cells.